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EDUCATION

PHYSICS OF FLUIDS GROUP, UNIVERSITY OF TWENTE

Jun. 2020 - Nov. 2023

Ph.D. IN FLUID PHYSICS (SUPERVISED BY: PROF. DETLEF LOHSE, PROF. ROBERTO VERZICCO)

- Turbulent flows with ice melting and freezing dynamics; → Pub [9]–[11], [14]–[18], [27]
- Respiratory droplet spreading and indoor ventilation; → Pub [12], [13], [22], [24], [25]
- With highest honors (Cum Laude) distinction.

UNIVERSITY OF TWENTE & MAX PLANCK INSTITUTE DYNAMICS AND SELF-ORGANIZATION

Sept. 2018 - May 2020

M.S. IN FLUID PHYSICS (SUPERVISED BY: PROF. DETLEF LOHSE, PD DR. OLGA SHISHKINA)

- Heat transfer enhancement of periodically modulated Rayleigh-Bénard convection; → Pub [26]
- Grade 9.1/10 with highest honors (Cum Laude) distinction.

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

Sept. 2014 - Jun. 2018

B.A. IN MECHANICS (SUPERVISED BY: PROF. ERQIANG LI, PROF. HANG DING)

- Compound droplet impact on dry surface.

EXPERIENCE

PRINCETON UNIVERSITY

Oct. 2024 - now / USA

POSTDOCTORAL RESEARCH ASSOCIATE (SUPERVISED BY: PROF. LUC DEIKE)

- Ocean waves modeling. → Pub [2] and [30]

PHYSICS OF FLUIDS GROUP

Dec. 2023 - Sep. 2024 / Netherlands

POSTDOC RESEARCHER (SUPERVISED BY: PROF. DETLEF LOHSE, PROF. ROBERTO VERZICCO)

- The physics of coupling between phase transition and turbulent transport of heat. → Pub [5], [6], [9]

WHOI GEOPHYSICAL FLUID DYNAMICS (GFD) PROGRAM

Jun. 2022 - Sep. 2022 / USA

GFD FELLOW (SUPERVISED BY: PROF. WANYING KANG, PROF. GLENN FLIERL)

- Numerical modeling of rotating equatorial flow with ice topography on Enceladus. → Pub [19]

MAX PLANCK INSTITUTE FOR DYNAMICS & SELF-ORGANIZATION

Jun. 2019 - May 2020 / Germany

RESEARCH INTERN (SUPERVISED BY: PD DR. OLGA SHISHKINA)

- Data-driven method for identification of the spatiotemporal structures. → Pub [23]

SKILLS

RESEARCH FIELDS Multiphase Flows | Turbulent Convection

NUMERICAL SKILLS Direct Numerical Simulation | Phase-Field Method | Data-Driven Modeling

PROGRAMMING Python | Fortran

LANGUAGES English | Chinese

ACADEMIC SERVICE

[1] Early Career Editorial Board Member, Theoretical and Applied Mechanics Letters, 2025-2030

[2] Reviewer for: Phys. Rev. Lett., Nat. Commun. Phys., J. Fluid Mech., Phys. Fluids, Phys. Rev. Fluids, Int. J. Multiph. Flow, Int. J. Heat Mass Transf.

HONORS & AWARDS

- | | |
|--|------|
| • Ehrenfest-Afanassjewski thesis award (Honourable mention) by Dutch Physics Council | 2024 |
| • ECCOMAS Best Thesis nominees (the only winner from the Netherlands) | 2023 |
| • Research highlight by 2023 JMBC Burgerscentrum | 2023 |
| • PhD graduate with highest distinctions (Cum laude) | 2023 |
| • WHOI Geophysical Fluid Dynamics (GFD) Fellowship | 2022 |
| • Master graduate with highest distinctions (Cum laude) | 2020 |

• University of Twente Scholarship	2018-2020
• Tsien Hsue-Shen Mechanical Talent Program	2015-2018
• National Endeavor Fellowship (Top 5%)	2016,2017
• Outstanding Student Gold Scholarship (Top 10%)	2014,2015

PUBLICATIONS

Published:

- [1] G. Vacca, **Rui Yang**, C. Howland, R. Verzicco, and D. Lohse, "Thermohaline circulation in a box": Heat and salinity transport in double-diffusive horizontal convection, *J. Fluid Mech.* 1034, A24 (2026) [16 pages].
- [2] **Rui Yang**, Z. Liu, P. K. Farsoiya, S. Popinet, and L. Deike, Surfactant effects on gravity-capillary waves, *J. Fluid Mech.* 1031, A2 (2026) [29 pages].
- [3] C. Demirkır, **Rui Yang**, A. Bashkatov, V. Sanjay, D. Lohse, and D. Krug, To jump or not to jump: Adhesion and viscous dissipation dictate the detachment of coalescing wall-attached bubbles, *Phys. Rev. Fluids* 10(12), 123602 (2025) [15 pages].
- [4] Z. Zheng, Y. Zhou, Y. Wang, Z. Cao, **Rui Yang**, Y. Li, E. Gu, J. Yao, Z. Wang, J. Ma, B. Yan, L. Shi High-value organic solvent recovery and reuse in perovskite solar cell manufacturing *Sci. Adv.* 11(34), eadt6008 (2025) [13 pages].
- [5] **Rui Yang**, D. Xu, R. Verzicco, and D. Lohse, Asymmetric melting and freezing in thermal convection, *J. Fluid Mech.* 1017, A12 (2025) [18 pages].
- [6] **Rui Yang**, and P. J. Schmid, Complex-network modeling of reversal events in turbulent thermal convection, *J. Fluid Mech.* 1011, A30 (2025) [18 pages].
- [7] X. Ye, K. Qin, **Rui Yang**, J. A. Dijkstra, Rebounding hydrogel sphere water entry, *J. Fluid Mech.* 1013, A2 (2025) [20 pages].
- [8] D. Xu, **Rui Yang**, R. Verzicco, D. Lohse, Aspect ratio effect on side and basal melting in fresh water, *J. Fluid Mech.* 1010, A40 (2025) [23 pages].
- [9] **Rui Yang**, C. Howland, H.-R. Liu, R. Verzicco, D. Lohse, Enhanced Efficiency of Latent Heat Energy Storage by Inclination, *PRX Energy* 3(4), 043006 (2024) [10 pages].
- [10] **Rui Yang**, K. L. Chong, H.-R. Liu, R. Verzicco, and D. Lohse, Melting and solidification in periodically modulated thermal convection, *J. Fluid Mech.* 998, A10 (2024) [17 pages].
- [11] **Rui Yang**, T. V. D. Ham, R. Verzicco, D. Lohse, and S. G. Huisman, Circular objects do not melt the slowest in water, *Phys. Rev. Fluids* 9(8), 083501 (2024) [13 pages].
- [12] G. Yerragolam, C. Howland, **Rui Yang**, R. J. A. M. Stevens, R. Verzicco, and D. Lohse, Effect of airflow rate on CO₂ concentration in downflow indoor ventilation, *Indoor Environ.* 1(2), 100012 (2024) [10 pages].
- [13] N. Leuenberger, **Rui Yang**, L. Munzel, R. Verzicco, D. Lohse, L. Bourouiba, and P. Jenny, Fast MILES solver: validation of a new scalable tool to study and optimize indoor ventilation, *Flow* 4, E23 (2024) [38 pages].
- [14] **Rui Yang**, C. Howland, H.-R. Liu, R. Verzicco, D. Lohse, Shape effect on solid melting in flowing liquid, *J. Fluid Mech.* 980, R1 (2024) [13 pages].
- [15] **Rui Yang**, C. Howland, H.-R. Liu, R. Verzicco, D. Lohse, Bistability in radiatively heated melt ponds, *Phys. Rev. Lett.* 131, 234002 (2023) [6 pages].
- [16] **Rui Yang**, C. Howland, H.-R. Liu, R. Verzicco, D. Lohse, Ice melting in salty water: layering and non-monotonic dependence on the mean salinity, *J. Fluid Mech.* 969, R2 (2023) [12 pages].

- [17] **Rui Yang**, C. Howland, H.-R. Liu, R. Verzicco, D. Lohse,
Morphology evolution of a melting solid layer above its melt heated from below,
J. Fluid Mech. 956, A23 (2023) [19 pages];
(see also the Cover of that issue).
- [18] **Rui Yang**, K. L. Chong, H.-R. Liu, R. Verzicco, D. Lohse,
Abrupt transition from slow to fast melting of ice,
Phys. Rev. Fluids 7(8), 083503 (2022) [12 pages];
- [19] **Rui Yang**, G. Flierl, W. Kang
Equatorial Ocean Dynamics on Enceladus Driven by Ice Topography,
WHOI GFD Proceeding volumes (2022) [24 pages];
- [20] H.-R. Liu, K. L. Chong, **Rui Yang**, R. Verzicco, and D. Lohse,
Turbulent Rayleigh-Bénard convection with bubbles attached to the plate,
J. Fluid Mech. 945, A29 (2022) [12 pages].
- [21] H.-R. Liu, K. L. Chong, **Rui Yang**, R. Verzicco, and D. Lohse,
Heat transfer in turbulent Rayleigh-Bénard convection within two immiscible fluid layers,
J. Fluid Mech. 938, A31 (2022) [15 pages];
(see also the Cover of that issue).
- [22] **Rui Yang**, C. S. Ng, K. L. Chong, R. Verzicco, and D. Lohse,
Do increased flow rates in displacement ventilation always lead to better results?
J. Fluid Mech. 932, A3 (2022) [12 pages];
(see also the Cover of that issue).
- [23] **Rui Yang**, X. Zhang, P. Reiter, M. Linkmann, D. Lohse, and O. Shishkina,
Data-driven identification of the spatio-temporal structure of turbulent flows by streaming Dynamic
Mode Decomposition,
GAMM-Mitteilungen, e202200003 (2022) [22 pages].
- [24] C. S. Ng, K. L. Chong, **Rui Yang**, M. Li, R. Verzicco, and D. Lohse,
Growth of respiratory droplets in cold and humid air,
Phys. Rev. Fluids 6, 054303 (2021) [15 pages].
- [25] K. L. Chong, C. S. Ng, N. Hori, **Rui Yang**, R. Verzicco, and D. Lohse,
Extended lifetime of respiratory droplets in a turbulent vapor puff and its implications on airborne dis-
ease transmission,
Phys. Rev. Lett. 126, 034502 (2021) [8 pages];
(see also the Cover of that issue).
- [26] **Rui Yang**, K. L. Chong, Q. Wang, R. Verzicco, O. Shishkina, and D. Lohse,
Periodically modulated thermal convection,
Phys. Rev. Lett. 125, 154502 (2020) [6 pages].
- [27] K. L. Chong, **Rui Yang**, Q. Wang, R. Verzicco, and D. Lohse,
Café Latte: spontaneous layer formation in laterally cooled double diffusive convection,
J. Fluid Mech. 900, R6, (2020) [12 pages].
- [28] S. Li, **Rui Yang**, K. Mu, X. Luo, and T. Si,
Thermal effects on the instability of coaxial liquid jets in the core of a gas stream,
Phys. Fluids 31, 032106 (2019) [15 pages].
- [29] S. Li, **Rui Yang**, X. Luo, and T. Si,
Instability study of an electrified coaxial jet in a coflowing gas stream,
Chin. J. Theor. Appl. Mech. 49(5), 997-1007 (2017).
- Ongoing:**
- [30] **Rui Yang**, J. Wu, N. Scapin, J. T. Farrar, B. Chapron, S. Popinet, and L. Deike,
Multilayer simulation of wind-forced breaking wave fields
(under review for *J. Fluid Mech.*).

TALKS

Invited talks:

1. 2024 9th ECCOMAS Congress, Lisboa, Portugal
2. 2024 Peking University, College of Engineering, Beijing, China
3. 2024 Lanzhou University, Lanzhou, China
4. 2023 University of Science and Technology of China, Hefei, China

5. 2023 JMBC Multiphase flow and phase transitions, Enschede, the Netherlands

Conference talks:

6. 2025 78th Annual meeting of APS-DFD, Houston, USA
7. 2025 Waves in the Sea Environment (WISE) 2025 – Seattle, USA
8. 2025 NASA International Ocean Vector Winds Science Team Meeting, Darmstadt, Germany (online)
9. 2024 International Conference on Numerical Methods in Multiphase Flows – 5, Reykjavik, Iceland
10. 2023 9th International Conference on Rayleigh-Bénard Turbulence, Xi'an China
11. 2023 18th European Turbulence Conference, Valencia, Spain
12. 2023 Closing workshop of DFG Priority Programme SPP 1881, Seeheim, Germany
13. 2022 14th European Fluid Mechanics Conference, Athens, Greece
14. 2022 Max Planck Center Meeting, Enschede, The Netherlands
15. 2022 Physics@Veldhoven, Veldhoven, The Netherlands (online)
16. 2021 Circular No. 20 meeting of DFG Priority Programme SPP 1881, Dresden, Germany
17. 2021 91st GAMM Annual Meeting (Mar 2021), Kassel, Germany (online)
18. 2020 73rd Annual meeting of APS-DFD, Chicago, USA (online)
19. 2020 Max Planck Center Meeting at Max Planck Institute for Polymer Research, Mainz, Germany
20. 2019 Summer School CMiF, "Complex Motion in Fluids", Kysthusene Gilleleje, Denmark